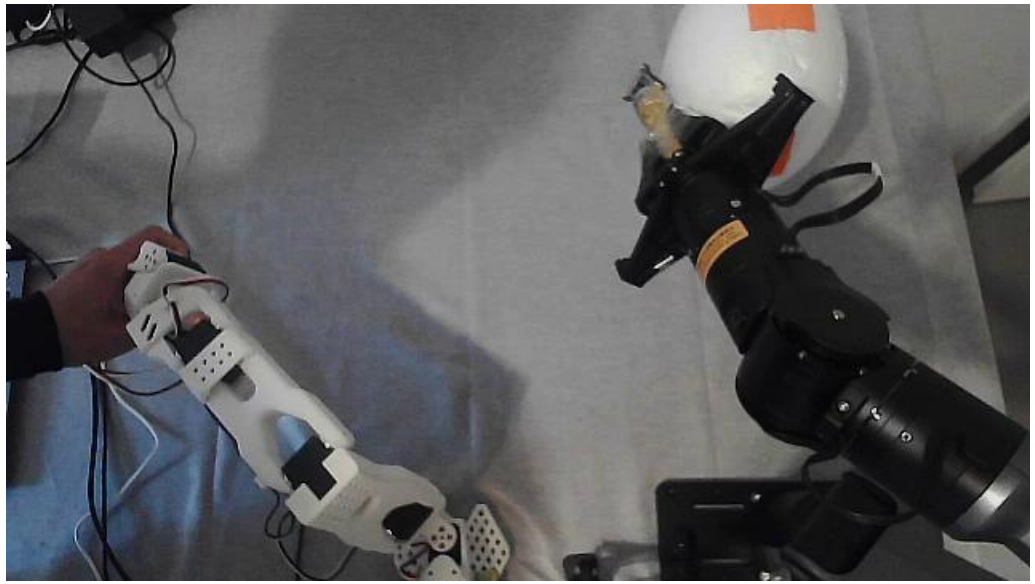
A Quadratic Programming Based Teleoperation Scheme and Its Applications for Robotic Haircutting

The slave follows the operator's pose within its joint limits, without joint correspondence



SO101 master arm (left) and Piper slave arm (right)

Outline

01 **Background and Motivation**
Why teleoperation for robotic haircutting

02 **Problem Formulation**
Pose tracking as a QP with joint limits

03 **Projection Neural Network**
Discrete-time solver, one step per instant

04 **Convergence Analysis**
Optimality, linear rate, joint limits

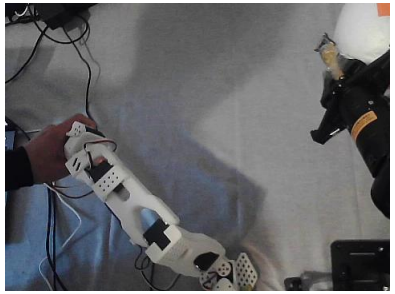
05 **Haircutting Experiments**
Master-arm mode and smartphone mode

06 **Conclusion**
Summary and future work

Motivation

Teleoperation keeps a human in the loop for haircutting, but joint-space mapping needs matching arms and a dedicated master.

Joint-space teleoperation



master arm

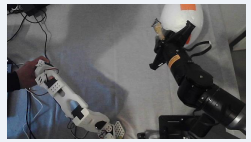
Joint-by-joint mapping



slave arm + mannequin head

- ? Different joint arrangements?
- ? Slave joint limits?
- ? Dedicated master arm?

Ours





master arm or smartphone

Desired pose →

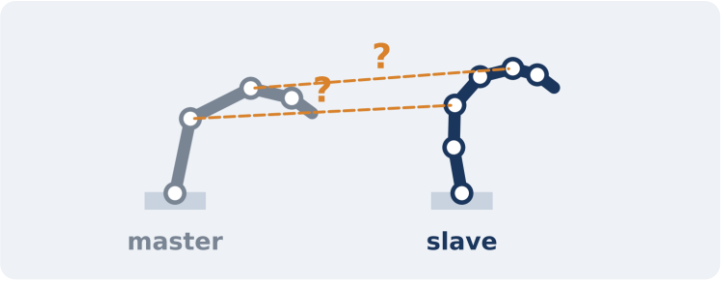
QP + projection neural network



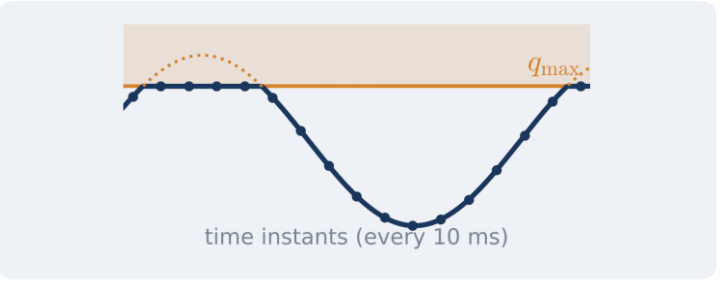
- ✓ No joint correspondence
- ✓ Joint limits at every instant
- ✓ Smartphone as input

Why Teleoperation for Haircutting Is Hard

The command must be realized by a different arm, near the head, within every sampling interval.







1 Heterogeneous arms

SO101 ≠ Piper

different joint arrangements and workspaces

The operator's command cannot be mapped joint by joint.

2 Dedicated input device

1 arm / site

a dedicated master arm at each operator site

The equipment cost limits access to remote haircutting.

3 Joint limits in real time

10 ms

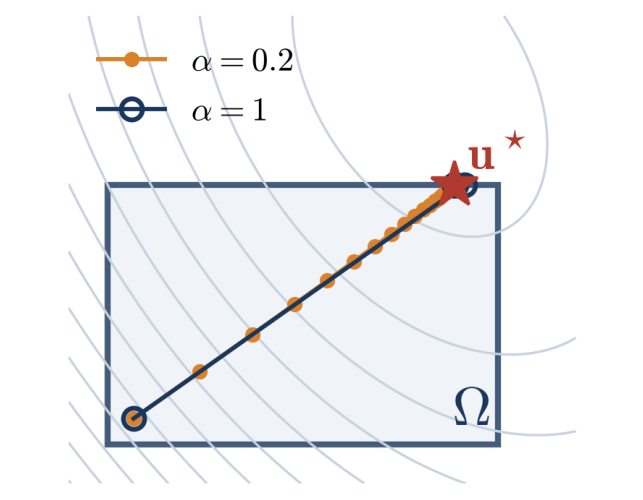
per time instant at 100 Hz

Every joint command has to stay within the angle and velocity limits.

What we need: a pose-level command that the slave realizes within its own joint limits, computed online, from any pose source.

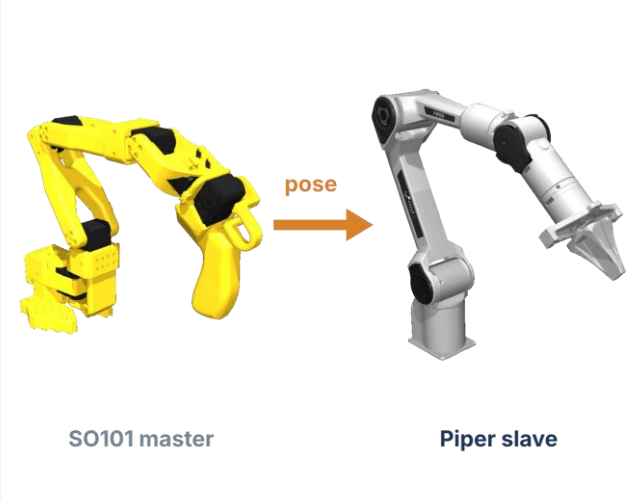
Our Contributions

To our knowledge, the first QP-based teleoperation scheme for robotic haircutting.



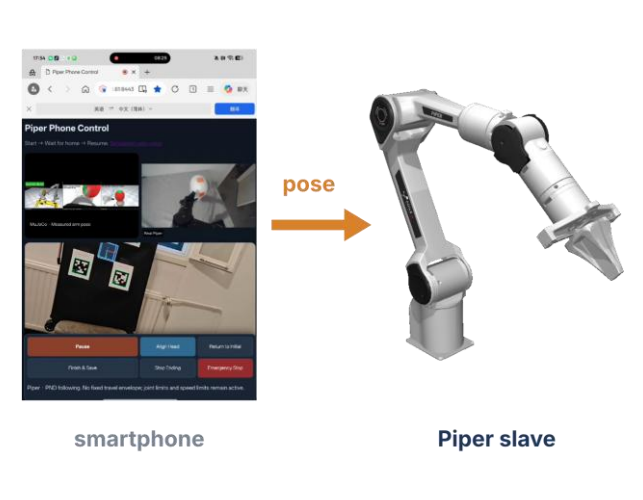
1 QP-based scheme

Pose tracking within joint limits, solved online by a discrete-time projection neural network.



2 Heterogeneous arms

An SO101 master teleoperates a Piper slave without joint correspondence.



3 Smartphone mode

A smartphone observing AprilTags replaces the master arm at low cost.

Preliminaries: Projection Neural Network

A projection neural network solves a convex problem over a box; the projection simply clips each element.

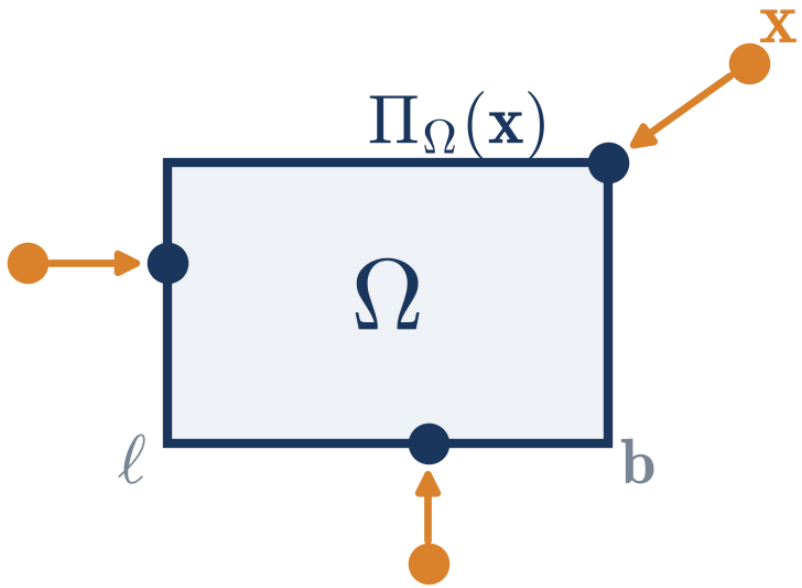
Projection neural network [Xia et al., 2002]

$$\epsilon \frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = -\mathbf{u} + \Pi_{\Omega}(\mathbf{u} - \gamma \nabla F(\mathbf{u}))$$

Projection onto the box Ω = {u | ℓ ≤ u ≤ b}

$$\Pi_{\Omega}(\mathbf{x}) = \max\left\{\min\left\{\mathbf{x}, \mathbf{b}\right\}, \ell\right\}$$

- F: convex objective; ε > 0: time constant; γ > 0: scaling parameter
- The state is stable at the optimal solution of min F(u) over Ω
- Elementwise clipping: cheap enough for every control cycle



Points outside Ω are moved to the nearest point of the box

Problem: Pose Tracking within Joint Limits

At each time instant, the joint velocity trades tracking accuracy against smooth motion, inside the joint limits.

Setting



Piper slave (6 joints) with a tool and a mannequin head

Desired pose: 6-D, from the master arm or the smartphone

No redundancy: 6 joints for a 6-D pose, so accuracy and small joint velocities conflict

1 Tracking error of the slave

$$\mathbf{e}_k = \mathbf{r}_{d,\,k} - \mathbf{r}(\mathbf{q}_{s,\,k})$$

2 Optimization problem for the joint velocity

$$\min_{\mathbf{u}_k} \ \tfrac{1}{2}\|\mathbf{W}(\mathbf{J}_k\mathbf{u}_k - k_e\mathbf{e}_k)\|^2 + \tfrac{\lambda}{2}\|\mathbf{u}_k\|^2 \quad \text{s. t.} \quad \ell_k \le \mathbf{u}_k \le \mathbf{b}_k$$

tracking term + **smoothness** term; a larger λ gives smoother motion and a larger error

3 Bounds from joint-angle and joint-velocity limits

$$\ell_k = \max\left\{-\mathbf{v}_{\max},\ (\mathbf{q}_{\min} - \mathbf{q}_{s,\,k})/h_k\right\}$$

$$\mathbf{b}_k = \min\left\{\mathbf{v}_{\max},\ (\mathbf{q}_{\max} - \mathbf{q}_{s,\,k})/h_k\right\}$$

Discrete-Time Projection Neural Network

Three steps turn the QP into a network that runs one step at each time instant.

Step 1 Optimality condition

$$\mathbf{u} = \Pi_{\Omega_k}(\mathbf{u} - \gamma \nabla F_k(\mathbf{u}))$$

Step 2 Continuous-time model

$$\epsilon \frac{\mathrm{d}\mathbf{u}}{\mathrm{d}t} = -\mathbf{u} + \Pi_{\Omega_k}(\mathbf{u} - \gamma \nabla F_k(\mathbf{u}))$$

Step 3 Euler forward, α = τ/ε

Discretize the model with time step τ, one step per control cycle (digital controller)

Network at the k-th time instant

$$\mathbf{u}_k = \mathbf{u}_{k-1} + \alpha[-\mathbf{u}_{k-1} + \Pi_{\Omega_k}(\mathbf{u}_{k-1} - \gamma \nabla F_k(\mathbf{u}_{k-1}))]$$

$$\mathbf{q}_{\mathrm{cmd},\,k} = \mathbf{q}_{s,\,k} + h_k \mathbf{u}_k$$

joint command

Compute e_k, J_k, H_k, f_k and the bounds → Project u_{k-1} onto Ω_k → **One network step → u_k** → Send $q_{cmd,k}$ to the slave

Smaller α: the command changes more smoothly (low-pass filter with inertia 1 − α), but approaches the optimum more slowly.

Convergence Analysis

The equilibrium is optimal, the error contracts linearly, and the joint limits hold at every instant.

Theorem 1 Optimal equilibrium

$$\bar{\mathbf{u}}_k = \mathbf{u}_k^\star \quad (\text{unique})$$

H_k is positive definite, so the QP has one optimum, and it is the equilibrium of the network.

Lemma 1 Non-expansive projection

$$\|\Pi_{\Omega_k}(\mathbf{y}) - \Pi_{\Omega_k}(\mathbf{x})\| \leq \|\mathbf{y} - \mathbf{x}\|$$

Projection onto the box never increases distances.

Theorem 2 Linear convergence

$$\|\mathbf{u}_k - \mathbf{u}_k^\star\| \leq \rho_k \|\mathbf{u}_{k-1} - \mathbf{u}_k^\star\|, \quad \rho_k = 1 - \alpha(1 - c_k)$$

If $0 < \alpha \leq 1$ and $0 < \gamma < 2/\bar{\sigma}_k$, then $\rho_k < 1$; for a fixed QP, convergence is global and linear.

Theorem 3 Joint limits at every instant

$$\mathbf{q}_{\min} \leq \mathbf{q}_{\text{cmd},\,k} \leq \mathbf{q}_{\max}\,, \qquad -\mathbf{v}_{\max} \leq \mathbf{u}_k \leq \mathbf{v}_{\max}$$

Holds whether or not u_k has reached the optimum, i.e., even with one iteration.

Accuracy bound: Theorem 3 also bounds the distance to the optimum by a residual that can be computed online at every time instant.

What the Theory Means in Practice

The error shrinks linearly, and the joint limits hold even before convergence.

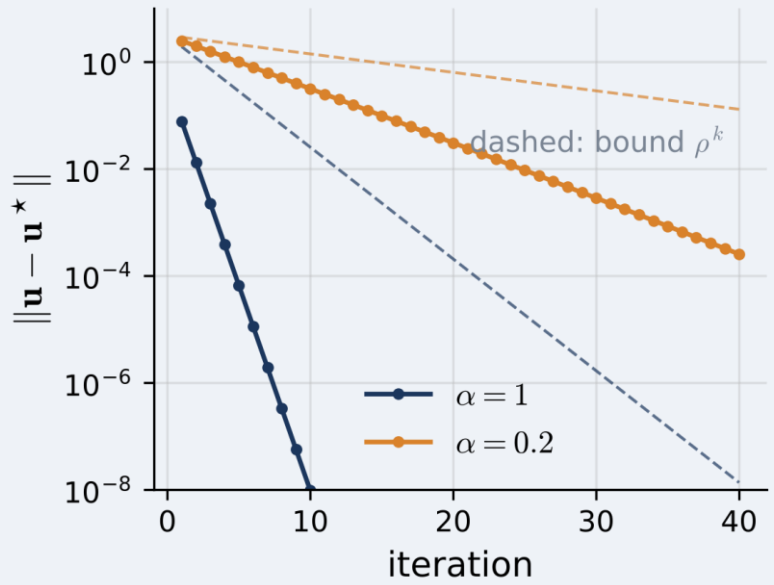
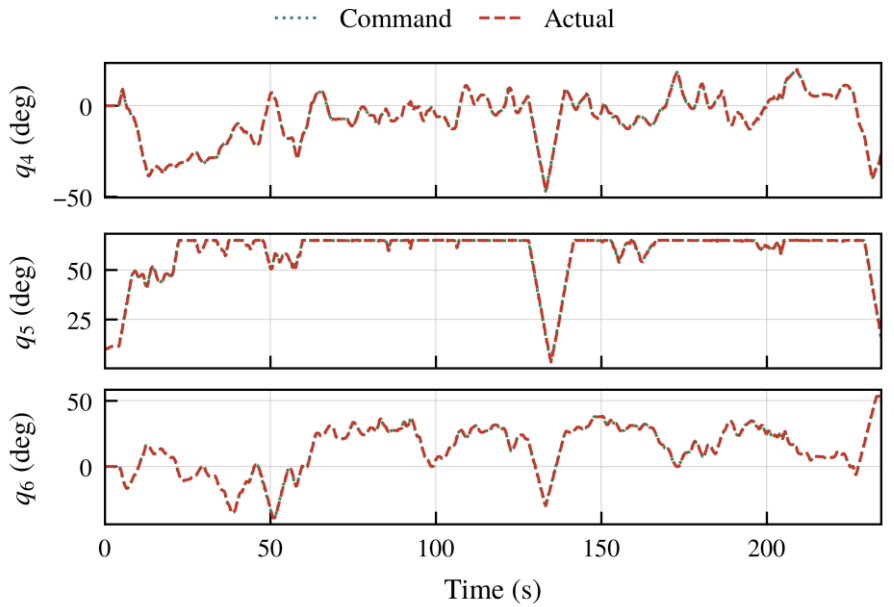


Illustration on a fixed 2-D QP (not from the paper): the distance to the optimum decays linearly on a log scale, below the bound ρ^k; $\alpha = 0.2$ is slower but smoother



Commanded and measured angles of joints 4–6, master-arm mode, α = 1 (paper figure)

65°

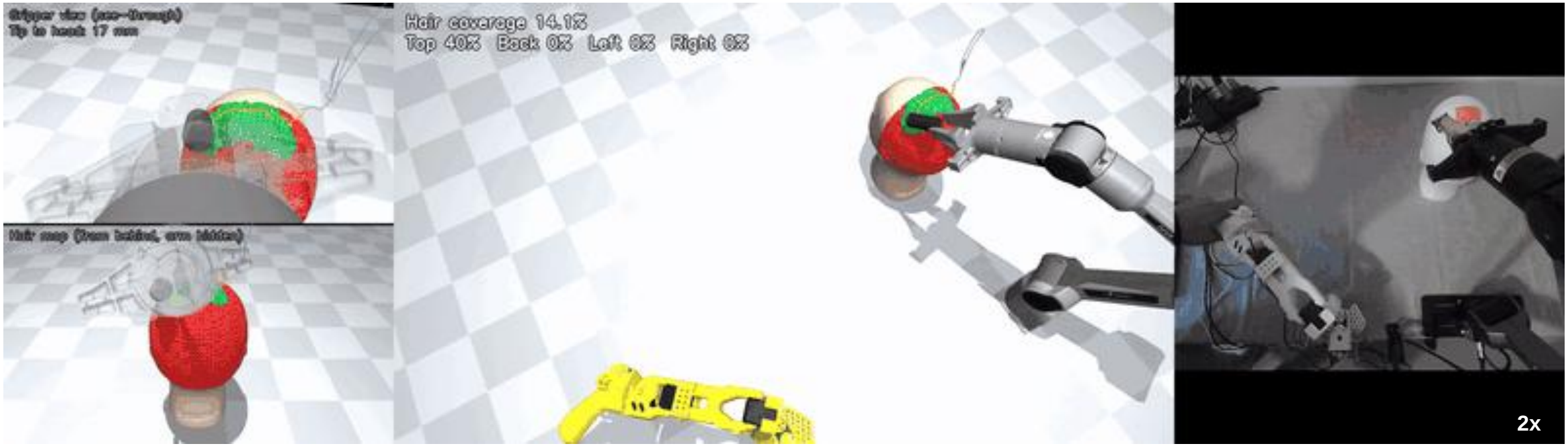
upper limit of joint 5, where it often rests

0

commanded angles outside the limits in both master-arm trials (Theorem 3)

Application 1: Heterogeneous Master–Slave Teleoperation

An SO101 master arm teleoperates a Piper slave arm; the master's end-effector pose is the desired pose.



Simulation views with model-based coverage (left, center) and camera view of the real arms (right)

Arms

SO101 master, Piper slave (6 joints)

Control

100 Hz, joint velocity ≤ 90°/s

Parameters

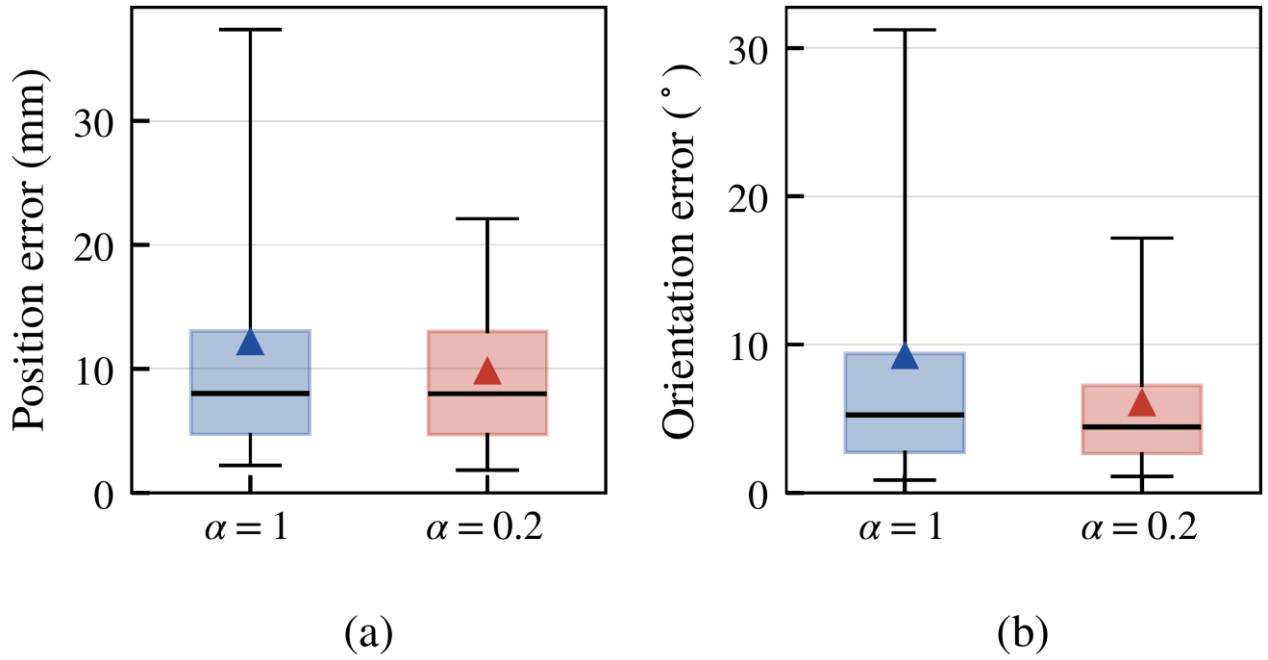
$w_p = 1$, $w_r = 0.1$, $k_e = 30$, $\lambda = 0.001$, $\gamma = 1/\bar{\sigma}_k$

Trials

$\alpha = 1$ (234.6 s) and $\alpha = 0.2$ (290.8 s)

Application 1: Tracking Errors

α = 0.2 lowers the mean and peak errors, while the median position error is almost unchanged.



Feedback tracking errors: boxes are quartiles, lines medians, whiskers 5th–95th percentiles, triangles means

12.12 → 9.76 mm

mean position error, α = 1 → α = 0.2

9.20° → 6.01°

mean orientation error, α = 1 → α = 0.2

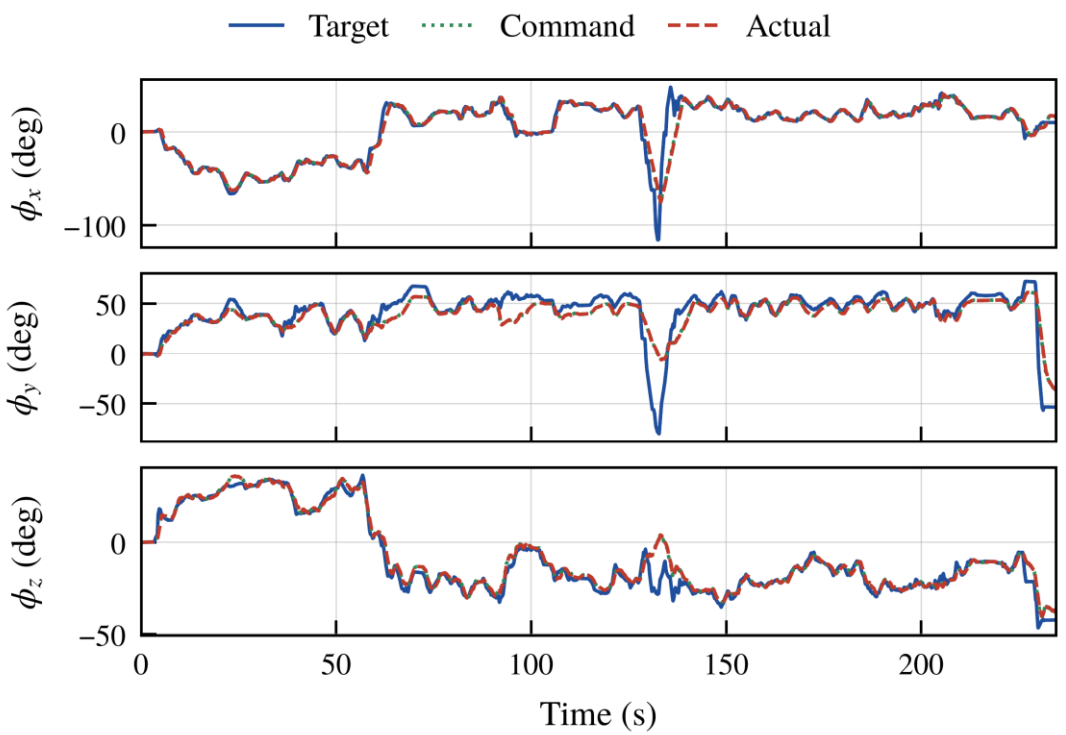
183.82 → 69.21 mm

maximum position error, α = 1 → α = 0.2

Where the large errors come from (α = 1): a wrist singularity from 129 s to 141 s, and the last 5 s, when the desired pose leaves the reachable region.

Application 1: Effect of Joint Limits

When joint 5 sits at its limit, the orientation yields while the position is still tracked.



Orientation tracking, α = 1: the error mainly occurs in φ_y (65–125 s)

Mode	Group	Samples	Pos. (mm)	Ori. (°)
Master, α = 1	Away	8260	17.70	12.45
	Near	14095	8.86	7.29
Master, α = 0.2	Away	19197	9.95	5.61
	Near	8515	9.34	6.91

Mean errors grouped by limit proximity (near: a commanded angle within 0.5° of its limit)

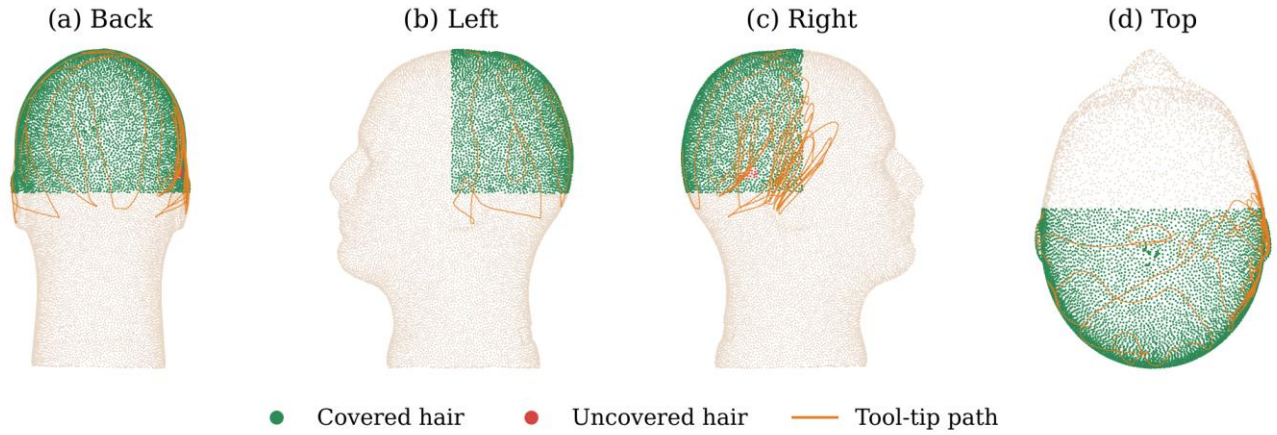
6.06° vs 3.41°

median orientation error with joint 5 at its upper limit vs otherwise (α = 1)

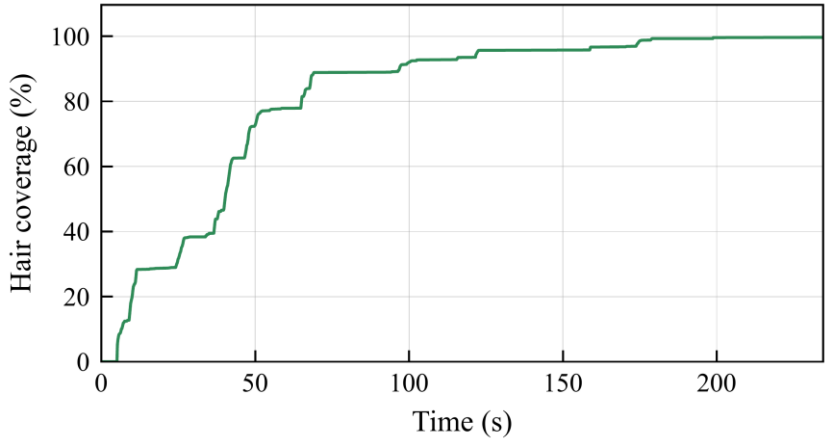
Orientation weight $w_r = 0.1$ < position weight $w_p = 1$, so the position is kept when both cannot be met.

Application 1: Coverage and Computation Time

The tool covers almost the whole modeled hair region, and one step takes well under the sampling interval.



Tool-tip path projected onto the head from four views (green: covered, red: uncovered), α = 1



Cumulative coverage, α = 1

99.7%

final geometric coverage, α = 1 (99.5% for α = 0.2)

0.72 ms

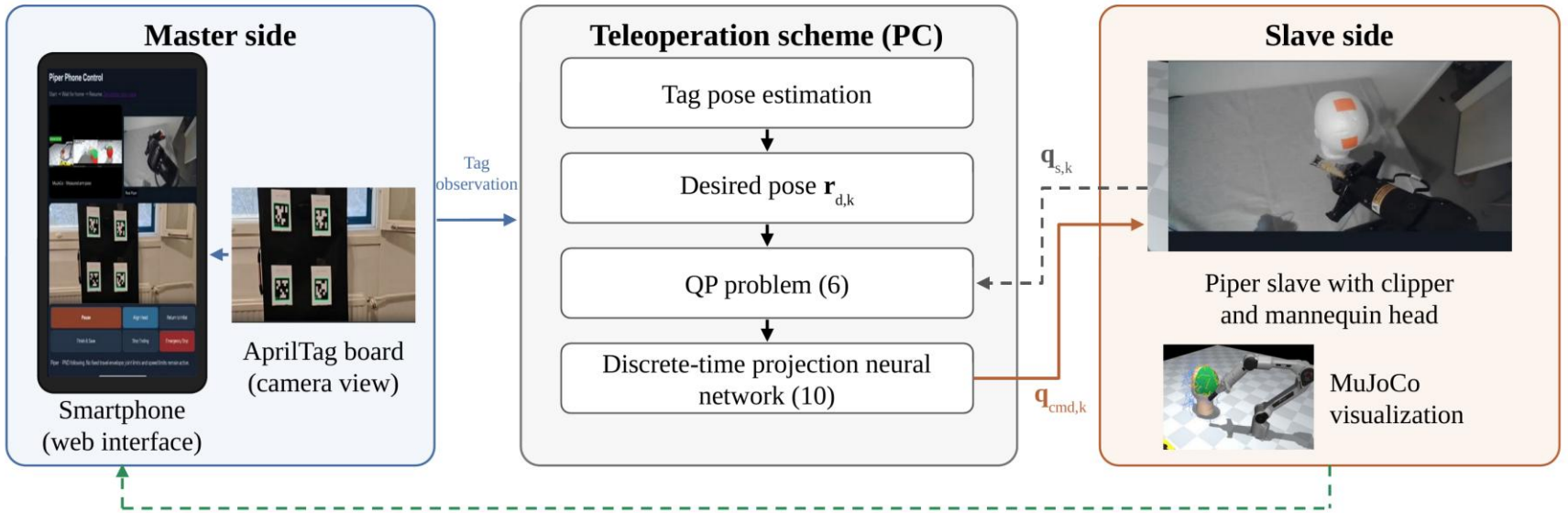
mean computation time per instant, both trials

10 ms

sampling interval at 100 Hz

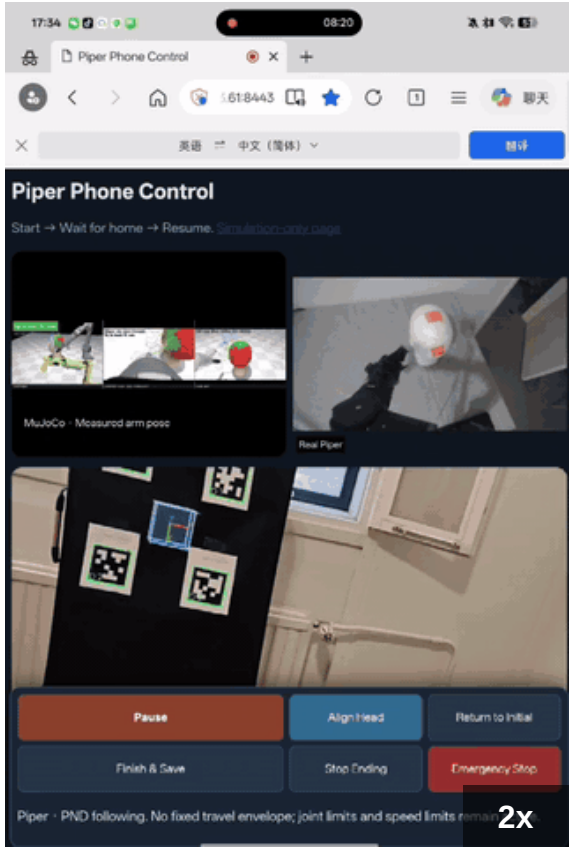
Application 2: Smartphone-Based Teleoperation via Tag

The smartphone camera observes a board with four AprilTags; the estimated pose is the desired pose of the slave.



Schematic of the smartphone mode (paper figure)

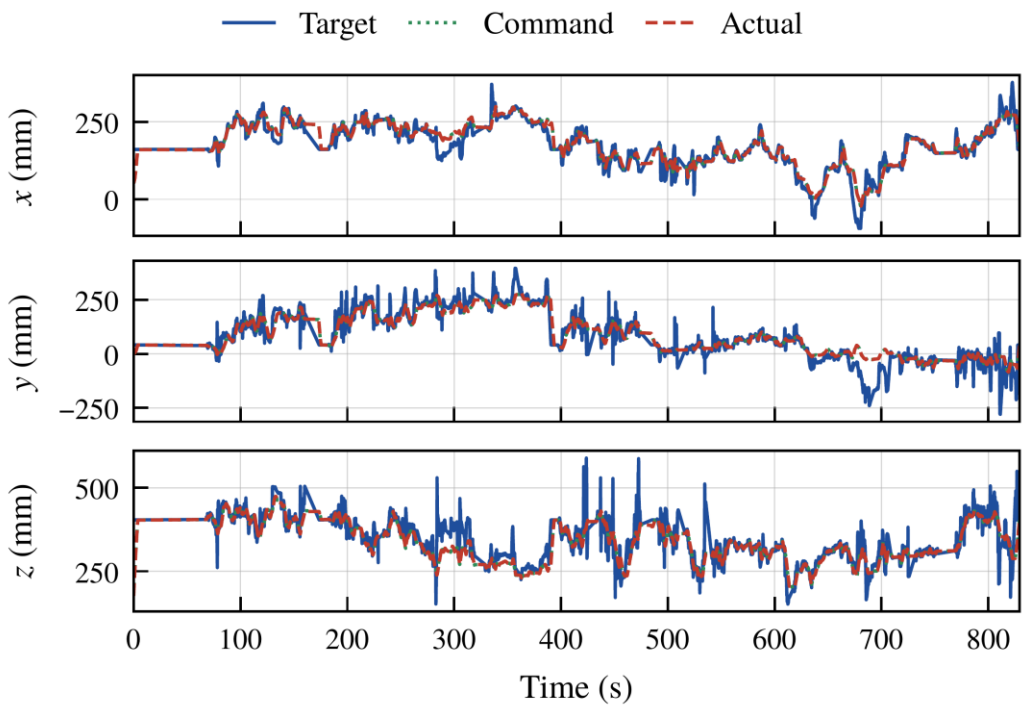
829.2 s trial · median sampling interval 24 ms · iterated to convergence at each instant



Smartphone screen: MuJoCo view and camera view of the slave

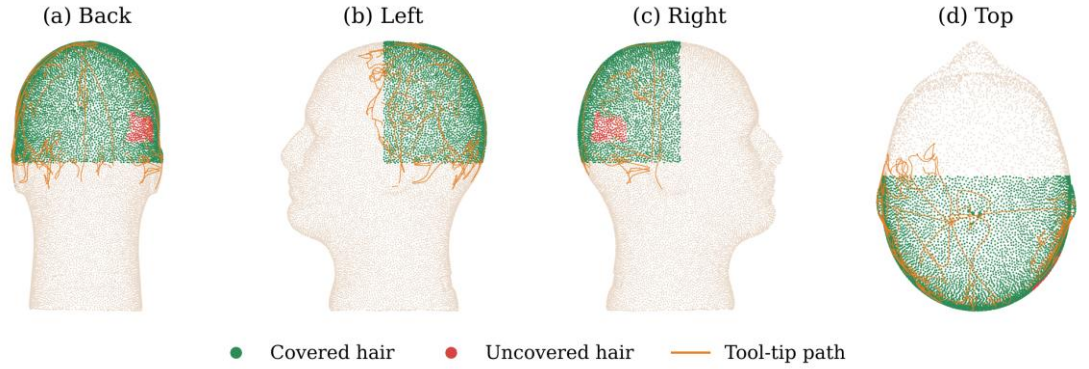
Application 2: Results

Both modes complete the task; the hand-held smartphone gives a noisier desired position.



Smartphone mode: desired (Target), command-derived and actual positions

Quantity	Master arm	Smartphone
Mean position error (mm)	12.12	49.67
Mean orientation error (°)	9.20	4.47
Mean computation time (ms)	0.72	1.88
Geometric coverage (%)	99.7	96.96



Smartphone mode: tool-tip path on the head (green: covered, red: uncovered)

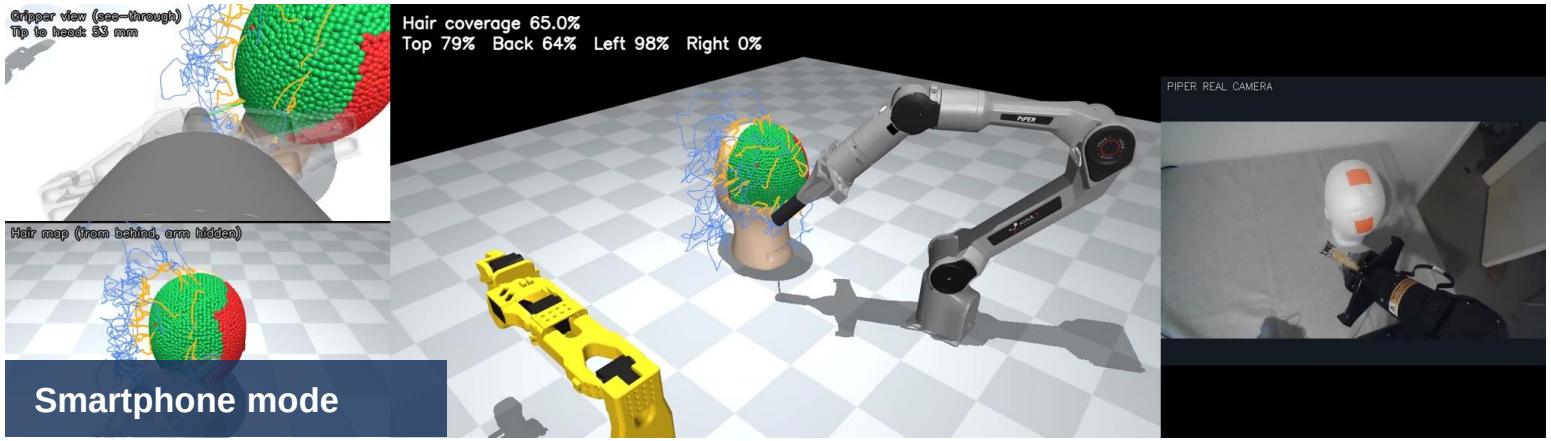
Joint-velocity limits smooth the rapid hand motion, which causes short peaks of the position error.

Conclusion

QP-Based Heterogeneous Teleoperation for Robotic Haircutting

an SO101 master with a Piper slave, and a smartphone as a low-cost master





Thank you